

Weichao Huang

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I'm interested in AI engineering because it offers the chance to turn new technology into tools that make everyday work easier. I'm especially curious about some research fields on AI agent, AI Infra, and LLM Inference.

EDUCATION

Johns Hopkins University

Master of Computer Science

Jan 2026 - May 2027

Baltimore, MD

Coursework: Machine Translation, Machine Programming, Parallel Programming, Probabilistic Models of the Visual Cortex, Computer Vision, Artificial Intelligence, NLP for Computational Social Science

University of Minnesota, Twin Cities

Bachelor of Computer Science

Jan 2024 - Jan 2026

Minneapolis, MN

Coursework: Operating Systems, Algorithms and Data Structures, Database Systems, Computer Networks, Distributed Systems, Computer Architecture and Organization, Software Engineering, Introduction to Machine Learning, Applied machine learning.

EXPERIENCES

Johns Hopkins University Medicine | Graduate Software Engineer

Mar 2026 - Jul 2026

DynaFace Mobile App Development

Baltimore, MD

- Built the iOS app in SwiftUI for an AI tool that scores facial palsy from a patient's video, used by patients and clinicians at Johns Hopkins University Medicine.
- Owned the full capture to result flow: guided video recording, upload, running the AI model in the cloud, and showing the marked up video and a 5 area facial function scorecard.
- Built account security end to end: email verification, password reset, account deletion, and a patient control that hides their data from any chosen clinician.
- Designed on a serverless Firebase and Google Cloud backend with async video jobs, built to serve a target of 1,000 to 2,000 patients and scale without managing servers.

PROJECTS

MiniMind | Python, PyTorch, LoRA, DPO, GRPO

Mar 2026 – Jun 2026

- Trained a 64M-parameter language model from scratch through pre-training, SFT, LoRA, DPO, and GRPO.
- Implemented and analyzed core Transformer components including RMSNorm, RoPE, and Grouped Query Attention (GQA).
- Fine-tuned the model for multi-turn conversation and evaluated differences across pre-training, SFT, and preference-alignment stages.
- Explored inference optimizations including KV caching and attention memory efficiency.

ChatMind - AI Agent System | Java, Spring AI, PostgreSQL

Dec 2025 - Feb 2026

- Built the core workflow of an AI agent using Spring AI, enabling it to call tools and complete tasks in multiple steps.
- Added knowledge-base search with Pgvector, allowing the agent to find relevant information and use it when answering questions.
- Added support for switching between AI models and used SSE to show live task updates, allowing users to follow the agent's progress.

SKILLS

- Programming Languages:** Python, Go, Java, C++, JavaScript, TypeScript
- Frameworks:** PyTorch, Spring Boot, Spring AI, Node.js, Django, Flask, Gin, Go-Zero
- Cloud & Infrastructure:** AWS, Docker, K8S, Kafka, CI/CD, Linux, Nginx
- Databases:** PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch
- AI & ML:** LLM Agent, RAG, Embedding, Prompt Engineering, SSE Streaming